

Exercise 4.11

Show that $\int_{-\infty}^{\infty} n(x) dx = 1$

$$n(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Solution

To show this we can start with simplifying the integral using substitution. We can perform a change of variables and scale the distribution to the standard normal form. This would be where $z = \frac{x-\mu}{\sigma}$ and $dz = \frac{1}{\sigma} dx$. After substituting we get the following:

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz$$

In order to solve the inner integral we need to use the gaussian integral trick which has $I = \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz$ and we square this term and then switch to polar coordinates:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx \right) \left(\int_{-\infty}^{\infty} e^{-\frac{1}{2}y^2} dy \right) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2+y^2)} dx dy$$

When we switch to polar coordinates we have $x^2 + y^2 = r^2$ and $dx dy = r dr d\theta$:

$$I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-\frac{1}{2}r^2} r dr d\theta$$

- **Inner Integral:** $\int_0^{\infty} e^{-\frac{1}{2}r^2} r dr$ with $u = \frac{1}{2}r^2$, $du = r dr$ will evaluate to $-e^{-u}|_0^{\infty} = 1$
- **Outer Integral:** $\int_0^{2\pi} 1 d\theta = 2\pi$

This gives us $I^2 = 2\pi$ or $I = \sqrt{2\pi}$. If we plug this back into the simplified integral in the first step we get the following:

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz = \frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$$

Exercise 4.18

Find the mathematical expectation of a random variable with

1. uniform distribution over the interval $[a, b]$
2. triangle distribution
3. exponential distribution

Solution

1. Uniform Distribution $[a, b]$

We can use the probability density function for the uniform distribution: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$.

1. Integral Setup:

$$E[X] = \int_a^b x \left(\frac{1}{b-a} \right)$$

2. Evaluate Integral:

$$E[X] = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right)$$

3. Simplify using difference of squares: $(b^2 - a^2) = (b-a)(b+a)$

$$E[X] = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}$$

2. Triangle Distribution

We can assume a symmetric triangle distribution on $[a, b]$ with the peak at $c = \frac{a+b}{2}$

1. Symmetry Argument: Since the distribution is symmetric around the peak at c , the center of mass must lie at that point.
2. Expectation Result:

$$E[X] = \frac{a + b + c}{3}$$

3. Exponential Distribution

The probability density function for this distribution is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

1. Integral Setup:

$$E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

2. Integration by Parts: We let $u = x$ and $dv = \lambda e^{-\lambda x} dx$. Then $du = dx$ and $v = -e^{-\lambda x}$.

$$E[X] = [-xe^{-\lambda x}]_0^\infty + \int_0^\infty e^{-\lambda x} dx$$

3. Evaluate Integrals:

- First Term: $[-xe^{-\lambda x}]_0^\infty$, this evaluates to 0 as $x \rightarrow \infty$.
- Second Term: $[-\frac{1}{\lambda}e^{-\lambda x}]_0^\infty = 0 - (-\frac{1}{\lambda}) = \frac{1}{\lambda}$.

4. Final Results:

$$E[X] = \frac{1}{\lambda}$$